

The Internal State Instrument

What the resolution operator measures, once resolution belongs to the body rather than to the observer

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WHAT THE INSTRUMENT READS

Addendum A01 fixed the resolution invariantly as $N = q / \delta q$. Read that with δq as measurement error and the operator describes the observer. Read it with δq as the quantity's **own spread** and it describes the object.

An oblate rotating body does not have one surface gravity. Its equator and its pole genuinely differ, so g has no single value to know. That spread is not ignorance, it is the body, and it fixes N before anyone measures anything.

The reading it produces is decoupled from the strength of the gravity. A millisecond pulsar and Saturn sit within a factor of two in N while their compactness differs by seven orders of magnitude. N measures how sharply a body defines a quantity, not how much of that quantity it has.

Every g below is computed from mass, the two radii and the rotation period. No value of surface gravity is looked up.

SECTION ONE

The two states

External is the number that gets quoted. Internal is what survives the body's own spread.

For a body of mass M with equatorial radius R_e , polar radius R_p and angular rate ω , the two extremes of surface gravity follow from the definition:

$$g_{\text{pol}} = GM / R_p^2 \quad g_{\text{eq}} = GM / R_e^2 - \omega^2 R_e$$

This omits the quadrupole contribution to the potential, so it lands within about a third of a per cent of published equatorial and polar values. It is the right approximation because the spread is dominated by the two radii and the centrifugal term, and both are present.

The external state is the midpoint of that spread and the half-range is δg . Centring anywhere else makes N depend on a convention rather than on the body, and shifts it by a few per cent:

$$g_{\text{ext}} = (g_{\text{eq}} + g_{\text{pol}})/2 \quad \delta g = |g_{\text{pol}} - g_{\text{eq}}|/2 \quad N = g_{\text{ext}}/\delta g$$

The internal state is then the external state charged for its own spread, through the operator of A_01 :

$$g_{\text{int}} = g_{\text{ext}} \cdot H_5(N) \quad H_5(N) = 1 + \tanh(5 \ln [N/(N+1)])$$

SECTION TWO

Ten bodies

Ordered by their own resolution. The corpus floor sits at $40\phi^3 = 169.44$ and cuts between Mars and Earth.

Body	g external	g equator	g pole	N	Deficit	g internal	Side
Saturn	10.8307	8.8266	12.8349	5.40	69.049%	3.3522	below
Jupiter	25.4604	22.5758	28.3450	8.83	49.044%	12.9737	below
PSR B1937+21	1.207e+12	1.093e+12	1.321e+12	10.59	42.287%	6.968e+11	below
Uranus	8.9492	8.6076	9.2907	26.20	18.512%	7.2925	below
Neptune	11.1961	10.8553	11.5368	32.86	14.877%	9.5304	below
Mars	3.7268	3.6962	3.7574	121.82	4.086%	3.5746	below
Earth	9.8144	9.7644	9.8644	196.38	2.539%	9.5652	above
Moon	1.6240	1.6220	1.6260	801.75	0.623%	1.6139	above
Sirius B	4.093e+06	4.093e+06	4.094e+06	9666.17	0.052%	4.091e+06	above
Sun	274.2055	274.2025	274.2084	93533.53	0.005%	274.1908	above

The division is not arranged. **The gas giants and the millisecond pulsar fall below the join, the Moon and the white dwarf and the Sun fall above it, and the boundary lands between Mars and Earth.** Below the join the instrument makes a claim the corpus licenses; above it the reading is quieter than the framework's own floor and cannot be offered as one.

SECTION THREE

The decoupling

N is independent of gravitational strength. This is the whole point of the instrument.

If the reading were a disguised restatement of how much gravity a body has, it would order the table the same way compactness does. It does not. The two orderings are unrelated:

Body	Compactness $2GM/Rc^2$	Flattening	Rotation q	N
Saturn	1.473e-08	9.796e-02	1.548e-01	5.40
Jupiter	4.076e-08	6.487e-02	8.919e-02	8.83
PSR B1937+21	3.462e-01	1.250e-02	1.515e-01	10.59
Uranus	5.103e-09	2.293e-02	2.953e-02	26.20
Neptune	6.195e-09	1.708e-02	2.608e-02	32.86
Mars	2.815e-10	5.889e-03	4.595e-03	121.82
Earth	1.393e-09	3.353e-03	3.461e-03	196.38
Moon	6.278e-11	1.243e-03	7.592e-06	801.75
Sirius B	5.283e-04	1.034e-04	0.000e+00	9666.17
Sun	4.245e-06	0.000e+00	2.138e-05	93533.53

TWO OBJECTS THAT SETTLE IT

Sirius B, a white dwarf. Compactness 5.28×10^{-4} , some two thousand times the Earth's. It is also one of the sharpest bodies in the table, $N = 9666$, because it barely rotates and is very nearly spherical. Enormous gravity, almost no internal ambiguity.

PSR B1937+21, a millisecond pulsar turning at 642 hertz. Compactness 0.346, the most extreme entry by seven orders of magnitude. Its N is 10.6, **below the join**, alongside Jupiter. Enormous gravity, enormous internal ambiguity.

So the instrument is reading something gravity does not determine. That is the uncoupling: **an object's internal state is not recoverable from its external one**, and N is what separates them.

What N does track is shape and spin. Flattening and the rotational parameter both order the table nearly as N does, which is the expected result: a quantity is smeared across a body exactly to the extent the body fails to be a sphere at rest.

SECTION FOUR

Where the linear reading stops working

The deficit is $5\delta q/q$ only above the join. Below it the exact operator is required.

Ao1 gives the unresolved fraction as $5/N$ to leading order, which invites using $5\delta q/q$ everywhere. It does not hold everywhere:

Body	N	Exact deficit	Linear $5/N$	Error of the linear form
Saturn	5.40	69.0491%	92.5212%	+34.0%
Jupiter	8.83	49.0437%	56.6492%	+15.5%
PSR B1937+21	10.59	42.2874%	47.2170%	+11.7%
Uranus	26.20	18.5115%	19.0826%	+3.1%
Neptune	32.86	14.8774%	15.2156%	+2.3%
Mars	121.82	4.0855%	4.1045%	+0.5%
Earth	196.38	2.5391%	2.5461%	+0.3%
Moon	801.75	0.6232%	0.6236%	+0.1%
Sirius B	9666.17	0.0517%	0.0517%	+0.0%
Sun	93533.53	0.0053%	0.0053%	+0.0%

THE JOIN DOES TWO JOBS, NOT ONE

The linear form is good to better than half a per cent from Mars upward ($N = 122$) and fails badly below it, reaching a 26 per cent overstatement at Saturn.

So $N \dagger = 40\phi^3$ is both the resolution where the deficit drops below the corpus floor and, to within the width of one body, the resolution where the linear reading becomes trustworthy. Above the join you may use $5\delta q/q$ but the answer is sub-floor. Below the join the answer is claimable but you must use the exact operator. The two regimes are cleanly separated and the separation was not put there by hand.

SECTION FIVE

That it is genuinely unit-free

Jupiter's reading in five unit systems, including a deliberately absurd one.

The invariance is not a claim about care with units. It is structural: N is a ratio of two quantities that carry the same dimension, so no change of unit can touch it.

Declared unit	g external	δg	N	Deficit
m/s ²	25.4604	2.88462	8.82625	49.0437%
cm/s ²	2546.04	288.462	8.82625	49.0437%
ft/s ²	83.5316	9.46397	8.82625	49.0437%
g (Earth)	2.59627	0.294148	8.82625	49.0437%
furlong / fortnight ²	1.95043e7	2.20978e6	8.82625	49.0437%

The same instrument on the same body returns one number. Compare this with the decimal-lift form of the original formulation, where the Jupiter residual moved from -0.020 per cent to -19.2 per cent on a change of unit alone. That was the defect Ao1 removed, and this table is the check that it stayed removed.

SECTION SIX

Distances

The same instrument, wherever a separation genuinely fails to have one value.

Nothing in the construction is about gravity. It needs a quantity and a spread that belongs to the quantity rather than to the instrument, and separations supply both: an orbit has a perigee and an apogee, and a parallax has a formal error.

Separation	q (m)	δq (m)	N	Deficit	Side	What smears it
Earth to Moon	3.8440e+08	2.1000e+07	18.30	25.985%	below	perigee to apogee, a real excursion
1 AU	1.4960e+11	2.5000e+09	59.84	8.268%	below	Earth's own orbital eccentricity
Earth to Mars	2.2500e+11	1.0000e+11	2.25	95.066%	below	opposition to conjunction
Proxima Centauri	4.0175e+16	2.8000e+12	14348.21	0.035%	above	Gaia DR3 parallax
Galactic centre	2.5567e+20	9.4600e+17	270.26	1.846%	above	8.28 +- 0.031 kpc
Andromeda	2.3995e+22	6.6200e+20	36.25	13.524%	below	2.537 +- 0.070 Mly
Hubble length	1.3700e+26	1.1000e+25	12.45	36.804%	below	the Ho basins themselves

The Hubble length is below the join. Its spread is the disagreement between the two H_0 basins, which is the one entry in this table where the smearing is not a geometric excursion or a formal error but a genuine disagreement between two ways of asking. Addendum Ao3 treats that spread as structure rather than as error, and this reading is consistent with it: a quantity measured in two basins does not have one value, and its N says so.

SECTION SEVEN

What the instrument is good for

Three uses that follow, and one that does not.

Use	Status
Ranking bodies by definitional sharpness. The ordering by N is computable from mass, radii and period alone, needs no measurement of the quantity itself, and is unit-free.	PROVED. Section two.
Separating internal from external state. The gap $g_{\text{ext}} - g_{\text{int}}$ is fixed by the body and not by the observer, and is not recoverable from the gravitational strength.	PROVED as a construction. Section three.
Deciding whether a claim is licensed. N below $40\phi^3$ means the reading exceeds the corpus floor and can be argued; N above means it cannot.	PROVED. Sections two and four.
Predicting a measured value more accurately than general relativity.	OPEN, and not supported here. The instrument returns an internal state; nothing in this addendum identifies that state with an observable that has been measured. Section eight states what would.

SECTION EIGHT

What would test it

The instrument is a construction until an internal state is matched to something measured.

The honest position is that sections two through six establish a well-defined, unit-free, body-determined quantity, and do not establish that it corresponds to anything. The gap $g_{\text{ext}} - g_{\text{int}}$ for Jupiter is twelve and a half metres per second squared, which is not a small correction to a measurement, it is half the value. So the internal state is not a refined estimate of surface gravity and must not be presented as one.

Step	What it requires
1	Name the observable the internal state is supposed to be. A quantity that is half the external one is a different quantity, not a better measurement of the same one.
2	Find a body pair with nearly equal compactness and very different N , or the reverse. Section three supplies both. Any observable that tracks N and not compactness is a candidate.
3	Keep the instrument locked. The same δq rule, the same centring, across every body, declared before the comparison.
4	Restrict to N below 169.44. Above the join the framework's own floor is coarser than the effect and nothing can be concluded.

$N = q/\delta q$, with δq belonging to the object, measures how sharply a thing defines a quantity, and it is independent of how much of that quantity the thing has. Saturn is the least self-defined body in the table and the Sun is the most, by four orders of magnitude, and neither fact is visible in their gravity.

PROVENANCE

Addendum Ao4 to the Unified Mechanics corpus. Builds only on Ao1, which supplies the operator and the invariant definition of resolution. Every surface gravity is computed from mass, the two radii and the rotation period; masses and radii are standard published values and no value of g is used as input. Regenerates from `internal_state_instrument.py` in this folder. The corpus is not modified by this document.